

# WHY THE RIEMANN HYPOTHESIS RESISTS: A CLASSIFIER FOR ZERO–ABSENCE MECHANISMS, THE FINITE–TO–FULL POSITIVITY GAP, AND A FINITIZATION OBSTRUCTION UNDER LINEAR INDEPENDENCE

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ABSTRACT. We do not attempt to prove the Riemann Hypothesis (RH). We address the adjacent meta-question: *why does every formalized route to RH stall at the same place?* We introduce a four-condition classifier  $D_0$  for mechanisms that could force the absence of off-line zeros, validate it against roughly thirty historical programs (it reproduces the known outcome of each, including the one proven RH — function fields), and isolate the single discriminating clause: a candidate escapes only if the off-line bound is a *corollary of a general theorem* proven independently for a whole class the object inhabits, rather than a property special to the object (which, for  $\zeta$ , is RH itself). We then prove a structural statement: in each positivity family attached to  $\zeta$  — intersection/Hodge, moment/Hankel, total positivity/Pólya frequency, combinatorial Hodge — every *finite-order* positivity is unconditional while the *full-order* positivity equals RH, and the four hierarchies are the same gap. We identify the missing ingredient as a *finitization* of  $\zeta$ 's positivity, show that in the function-field case it is exactly a Frobenius periodicity, and prove that under the Linear Independence conjecture for the zeros no such periodicity — hence no finite-dimensional algebraic finitization — can exist. The geometric route to RH is therefore in tension with the believed independence of the zeros. None of this proves or disproves RH; it is a precise, conditional account of the obstruction, and it is falsifiable.

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## 1. INTRODUCTION

A century of work has produced many reformulations of the Riemann Hypothesis and several deep programs aimed at it, yet they stall. This paper is not one more reformulation. It asks a structural question — *what do the stalled programs have in common, and is that commonality forced?* — and answers it with three components: a classifier (§2), a structural theorem (§3), and a conditional obstruction (§4).

The classifier  $D_0$  is calibrated, not postulated: it is required to reproduce the historical outcome of every program it is run against. The one program it marks as a full success is the one proven RH (function fields, twice over, plus the cross-problem controls of modularity and Heegner points); every stalled program fails a specific, named clause. The decisive clause turns out to be sharp and non-circular, and it points to a single missing object, which the structural theorem and the obstruction then analyze.

**Standing disclaimer.** No statement below assumes RH, and none implies it. The results are about the *shape of the difficulty*.

2. THE CLASSIFIER  $D_0$ 

Fix the completed zeta function  $\xi$ , entire of order 1, with zeros  $\frac{1}{2} + i\gamma$  ( $\gamma \in \mathbb{R} \iff \text{RH}$ ). A *mechanism* is any mathematical structure  $X$  together with an asserted implication “[a property of  $X$ ]  $\Rightarrow$  [no zeros off the critical line, in some region].”

**Criterion 2.1** ( $D_0$ ). A mechanism  $X$  *escapes collapse* only if it satisfies all four:

- $I_1$  (*prime-distinguishing*) the asserted property depends on the individual primes (the Euler structure), not merely on prime counts or on  $\zeta$ ’s global analytic data;
- $I_2$  (*independent input with a decisive theorem*)  $X$  lies in a class  $\mathcal{C}$  carrying a general theorem  $T$ , established independently of RH and holding for *every* object of  $\mathcal{C}$ , such that  $T$  applied to  $X$  yields the off-line bound as a *corollary* ( $I_{2a}$ : such an independent  $\mathcal{C}, T$  exist;  $I_{2b}$ : the bound is a corollary, not a property special to  $X$ );
- $I_3$  (*individual-resolving*) the property constrains a single zero (an  $O(1)$  signal), not an aggregate sum, integral, moment, or density;
- $I_4$  (*arithmetic-aware*) the property depends on the actual primes and is not invariant under arithmetic-blind deformation.

*Remark 2.2* (The role of  $I_{2b}$ ).  $I_{2b}$  is the operative clause. For a surface  $C \times C$  over  $\mathbb{F}_q$ , the bound  $|\alpha_i| = \sqrt{q}$  is a corollary of the Hodge index theorem, a general fact about *all* algebraic surfaces; the object merely lands in the class where the theorem applies. For  $\zeta$ , the analogous positivity (the Weil pairing’s sign) is not a corollary of any general theorem proven elsewhere; it *is* RH. The distinction is a category-placement question — decided by exhibiting  $\mathcal{C}$  and  $T$  — not an inequality one must prove about  $\zeta$ ; Weil’s history (find the category, then the inequality follows from general geometry) is the proof of its non-triviality.

**Proposition 2.3** (Calibration). *Run against the standard corpus of programs — Weil’s explicit formula and criterion; Hilbert–Pólya and Berry–Keating; Connes’ trace formula; Deninger’s cohomology; Connes–Consani; de Branges canonical systems; Nyman–Beurling and Báez-Duarte; Li’s criterion; Montgomery pair correlation; random matrix theory and moment/ratio conjectures; multiplicative chaos; zero density estimates and Levinson–Conrey mollifiers; the de Bruijn–Newman flow; and the function-field proofs (Weil/Hodge and Stepanov) —  $D_0$  assigns to each a verdict in  $\{\text{PASS}, \text{OPEN}, \text{REJECT}\}$  that matches its historical status. In particular the only PASS verdicts are the proven cases (function fields; and, as cross-problem controls, modularity and Heegner-point/Gross–Zagier), and every REJECT fails a single named clause.*

We record the structure of the verdicts; the full table is in the supplementary audit. Three states arise, and the three-valuedness is essential: RECOGNIZE-PASS (a known general  $T$  applies for free: function fields); OPEN (a class is being built so a general  $T$  *would* apply, but  $T$  is not yet established: Connes–Consani, Deninger —  $I_{2a}$  holds,  $I_{2b}$  does not); REJECT (no general theorem is in sight; the bound is special to  $\zeta$ : all reformulations, all statistics, all flows).

*Remark 2.4* (The library is finite). The only general theorems that have ever delivered such a bound as a corollary are the Hodge-type positivity theorems: the Hodge index theorem (surfaces), Weil II/purity (varieties over finite fields), the Hodge–Riemann relations (Kähler; and, after Adiprasito–Huh–Katz, matroids and Lorentzian polynomials), the spectral theorem for a self-adjoint operator with independently known spectrum, and Stepanov’s auxiliary-function method. Running each against  $\zeta$  produces a single named obstruction per theorem, and the obstructions cluster into two roots: the absence of a geometry over  $\text{Spec } \mathbb{Z}$  richer than  $\zeta$  (no surface, no independent cohomology, no independent operator), and the object-special trap (any known embedding into a positivity class requires proving the positivity by hand, i.e. RH). The remainder of the paper analyzes these two roots and shows they are one.

### 3. THE FINITE-TO-FULL POSITIVITY GAP

We make precise the sense in which every positivity route to  $\zeta$  stops at the same place. Recall the de Bruijn–Newman kernel

$$\Phi(t) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9t/2} - 3\pi n^2 e^{5t/2}) e^{-\pi n^2 e^{2t}}, \quad \Phi > 0 \text{ unconditionally,}$$

with  $\widehat{\Phi} = \Xi$  (the standard normalization, so  $\Xi$  is real and even and  $\text{RH} \iff \Xi \in \text{LP}$ , the Laguerre–Pólya class). The Jensen/ $\Xi$  coefficients are the normalized moments  $b(k) = m_{2k}/(2k)!$ ,  $m_{2k} = \int t^{2k} \Phi(t) dt$ .

**Theorem 3.1** (Total-positivity form of RH). *RH holds if and only if  $\Phi$  is a Pólya frequency function, i.e. the kernel matrices  $[\Phi(x_i - x_j)]_{i,j=1}^N$  are totally positive (all minors  $\geq 0$ ) for all  $N$  and all  $x_1 < \dots < x_N$ .*

*Proof.* By Schoenberg’s theorem an integrable even kernel  $\Phi$  is a Pólya frequency function if and only if its bilateral Laplace transform is the reciprocal of a Laguerre–Pólya function. Since  $\widehat{\Phi} = \Xi$  and  $\text{RH} \iff \Xi \in \text{LP}$ , the two conditions coincide.  $\square$

Total positivity is a hierarchy  $\text{TP}_1 \supseteq \text{TP}_2 \supseteq \dots \supseteq \text{TP}_\infty$ , where  $\text{TP}_k$  requires all minors up to order  $k$  to be nonnegative. The following organizes the known unconditional results.

**Proposition 3.2** (The available orders). *For  $\zeta$ :  $\text{TP}_1$  ( $\Phi > 0$ ) is classical;  $\text{TP}_2$  (the Turán inequalities, equivalently log-concavity of  $b(k)$ , equivalently hyperbolicity of the degree-2 Jensen polynomials) holds unconditionally (Csordas–Norfolk–Varga); for each fixed  $k$ , the order- $k$  condition (hyperbolicity of the degree- $k$  Jensen polynomials) holds for all sufficiently large height (Griffin–Ono–Rolen–Zagier); and  $\text{TP}_\infty$  is equivalent to RH.*

**Theorem 3.3** (The finite-to-full gap). *Attach to  $\zeta$  the four positivity hierarchies: intersection/Hodge (the localized Weil form and its bottom eigenvalue), moment/Hankel (the  $m_{2k}$ ), total positivity (the kernel  $\Phi$ ), and combinatorial Hodge (the Jensen polynomials). In each, every finite order is unconditional for  $\zeta$ , the full order equals RH, and no general theorem of the family bridges a finite order to the full order for the specific object. Moreover the four hierarchies are mutually equivalent order by order: the second order is, in all four, the Turán/log-concavity/ $\text{TP}_2$ /degree-2-Jensen inequality, and the full order is, in all four,  $\Xi \in \text{LP}$ .*

*Proof sketch.* Equivalence of the full orders is the chain  $\lambda_{\min} \geq 0 \iff \Xi \in \text{LP} \iff \Phi$  Pólya frequency  $\iff$  all Jensen polynomials hyperbolic, each link classical (Weil criterion; Theorem 3.1; the Jensen–Pólya characterization of LP). Equivalence of the second orders is the identity of the Turán inequality  $b(k)^2 \geq b(k-1)b(k+1)$  with  $\text{TP}_2$  of  $\Phi$  and with the discriminant condition for the degree-2 Jensen polynomial. That every finite order is unconditional while the full order is RH is Proposition 3.2 in the total-positivity column and the cited results in the others. That no general theorem of a family bridges finite to full is the content of  $I_{2b}$ : the bridge is the object-special step, which is RH.  $\square$

*Remark 3.4.* Theorem 3.3 is the precise form of “every positivity route stops at the same place.” The obstruction is uniformity: order  $k$  is available for each  $k$  (eventually in height), but RH requires all orders at all heights. This is the cornered target reached from the total-positivity direction.

#### 4. FINITIZATION, AND THE OBSTRUCTION UNDER LINEAR INDEPENDENCE

We now explain why the function-field case has no finite-to-full gap, and what its absence would require for  $\zeta$ .

**Proposition 4.1** (Geometry finitizes positivity). *For a smooth projective curve  $C/\mathbb{F}_q$  of genus  $g$ , the full positivity that is RH for  $\zeta_C$  is a finite condition: the reality of the  $2g$  Frobenius arguments  $\theta_i$ , delivered by the Hodge index theorem on the finite-rank lattice  $\text{NS}(C \times C)$ . Equivalently, in the variable  $s$  ( $T = q^{-s}$ ) the zeros are*

$$s = \frac{1}{2} + i \frac{\theta_i + 2\pi k}{\log q} \quad (1 \leq i \leq 2g, k \in \mathbb{Z}) :$$

*infinitely many zeros, but only  $2g$  modulo the period  $2\pi/\log q$ .*

Thus the device that converts the infinite-order positivity into a finite one is a *periodicity*: the Frobenius folds the zeros into  $2g$  orbits. We call such a structure a *finitization*. By Theorem 3.3, a finitization is exactly what a proof of RH along any positivity route would require, and by §2 it is the only object in bin OPEN/NEW.

**Theorem 4.2** (Finitization obstruction under LI). *Assume the Linear Independence conjecture: the positive ordinates  $\{\gamma_n\}_{n \geq 1}$  of  $\zeta_{\mathbb{Q}}$  are linearly independent over  $\mathbb{Q}$ . Then there is no real period  $\tau > 0$  and finite set  $\{\beta_1, \dots, \beta_m\}$  with  $\{\gamma_n\} \subseteq \{\beta_j + k\tau : 1 \leq j \leq m, k \in \mathbb{Z}\}$ . In particular  $\{\gamma_n\}$  admits no Frobenius-type periodicity, hence no finite-dimensional algebraic finitization of the type of Proposition 4.1.*

*Proof.* If such  $\tau, \{\beta_j\}$  existed, then for each  $n$  there are  $j(n), k(n)$  with  $\gamma_n = \beta_{j(n)} + k(n)\tau$ . Taking two indices  $n \neq n'$  with  $j(n) = j(n')$  gives  $\gamma_n - \gamma_{n'} = (k(n) - k(n'))\tau$ , a rational multiple relation among the differences; pinning  $\tau$  from one such pair and substituting into a third index (there are infinitely many  $n$  but finitely many classes  $j$ , so some class is infinite) yields a nontrivial  $\mathbb{Q}$ -linear relation  $a\gamma_{n_1} + b\gamma_{n_2} + c\gamma_{n_3} = 0$  with  $(a, b, c) \in \mathbb{Q}^3 \setminus \{0\}$ , contradicting linear independence.  $\square$

**Corollary 4.3** (The  $\mathbb{F}_1$  degeneration). *The finitizing period of Proposition 4.1 is  $2\pi/\log q$ , which diverges as  $q \rightarrow 1^+$ . The fundamental domain holding finitely many zeros becomes infinite precisely in the  $\mathbb{F}_1$  limit; the finite structure dissolves where it is needed. The Connes–Consani program is thus an attempt to extract a finite residue from a divergence, consistent with its being stalled exactly at the arithmetic Hodge index /  $H^1$  term.*

*Remark 4.4* (The dichotomy). Theorem 4.2 forces a clean alternative. If LI holds (as is widely believed: simplicity of zeros, GUE spacing, absence of observed resonances), then no finite-dimensional algebraic finitization exists, and a positivity proof of RH would require an *infinite-dimensional* general positivity theorem; by  $D_0$  every known infinite-dimensional positivity for  $\zeta$  is object-special (the Weil/Krein form). If instead LI fails, a hidden periodicity might exist and the geometric route

reopens — but LI-falsity is itself a sensational and strongly disbelieved statement. Either way, the geometric route to RH is in tension with the independence of the zeros.

## 5. CONCLUSION

The three components combine to a single conditional statement. Every formalized positivity route to RH is, by Theorem 3.3, a climb up one of four equivalent finite-order hierarchies whose top is RH; crossing requires a finitization (§4); in the only setting where RH is proven the finitization is a Frobenius periodicity (Proposition 4.1); and under the Linear Independence conjecture no such periodicity exists for  $\zeta_{\mathbb{Q}}$  (Theorem 4.2), with the would-be  $\mathbb{F}_1$  substitute degenerating (Corollary 4.3). This is a precise, falsifiable account of why RH resists the one method that ever proved an RH — not a proof, and not a barrier theorem, but a conditional structural obstruction tying the difficulty of RH to the believed independence of its zeros. It is refuted the day a periodicity is exhibited (i.e. LI is disproved) or a general infinite-dimensional positivity theorem is found that applies to  $\zeta$  as a corollary.

**Acknowledgement of scope.** The classifier’s calibration is an expert judgment reproducing historical outcomes, not a theorem; the finite-to-full gap (Theorem 3.3) is a precise statement whose “no general theorem bridges” clause is the content of the open problem; the obstruction (Theorem 4.2) is conditional on LI. The contribution is the organization of these into one account and the identification of the single object — a finitization — whose existence or non-existence decides the geometric route.

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